

# TELCO CUSTOMER CHURN ANALYSIS

## Project Overview

The telecom industry has high customer churn, making it crucial to know why customers leave. This EDA helps to identify key patterns, trends and factors influencing customer churn by analysing customer demographics, services, account information and billing details. The goal is to uncover actionable insights that helps to improve customer retention and support future churn prediction models.

## Dataset Summary

-Rows: 7043

-Columns: 21

-Key Features:

- Customers who left within the last month – the column is called Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
- Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges
- Demographic info about customers – gender, age range, and if they have partners and dependents

-Missing Data: 11 missing values in TotalCharges columns

## Data Cleaning using Python

We began with data preparation and cleaning in Python:

- **Data loading:** Imported dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for statistical summary.

|   | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines    | InternetService | OnlineSecurity | ... | DeviceProtection | TechSupport |
|---|------------|--------|---------------|---------|------------|--------|--------------|------------------|-----------------|----------------|-----|------------------|-------------|
| 0 | 7590-VHVEG | Female | 0             | Yes     | No         | 1      | No           | No phone service | DSL             | No             | ... | No               | No          |
| 1 | 5575-GNVDE | Male   | 0             | No      | No         | 34     | Yes          | No               | DSL             | Yes            | ... | Yes              | No          |
| 2 | 3668-QPYBK | Male   | 0             | No      | No         | 2      | Yes          | No               | DSL             | Yes            | ... | No               | No          |
| 3 | 7795-CFOCW | Male   | 0             | No      | No         | 45     | No           | No phone service | DSL             | Yes            | ... | Yes              | Yes         |
| 4 | 9237-HQITU | Female | 0             | No      | No         | 2      | Yes          | No               | Fiber optic     | No             | ... | No               | No          |

5 rows × 21 columns

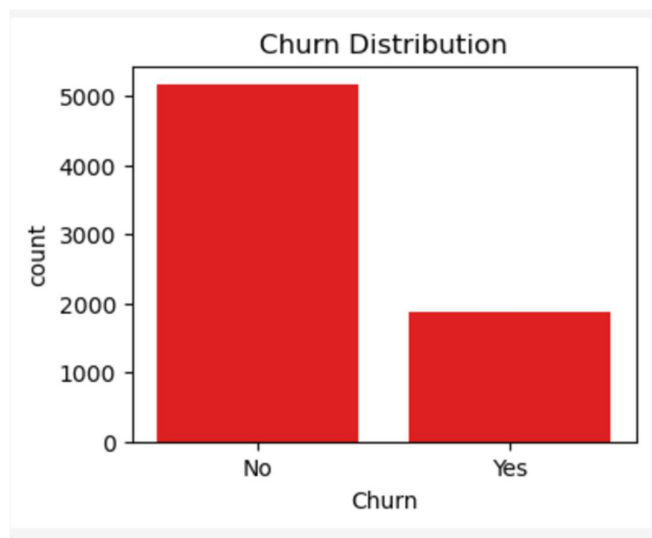
|   | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines    | InternetService | OnlineSecurity | ... | DeviceProtection | TechSupport |
|---|------------|--------|---------------|---------|------------|--------|--------------|------------------|-----------------|----------------|-----|------------------|-------------|
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5 rows × 21 columns

- **Fixed Data type:** Total charges data type (object) is converted into numerical data type.
- **Replace empty strings with NAN:** Total charges values are in float but are stored as string so we suspect that there are empty strings present in the data.
- **Missing Data Handling:** The distribution was positively skewed so we use median to replace the missing data in TotalCharges.
- **Outlier analysis:** The boxplot was used to observe the outliers in MonthlyCharges but the analysis show no observations outside whisker boundaries indicating the absence of significant outliers. Therefore, no outlier treatment was required for the this variable.
- **Feature Engineering:**
  - Created **TenureGroup** column with Tenure column.
  - Created **Spendinglevels** column from MonthlyCharges.

## Exploratory Data Analysis

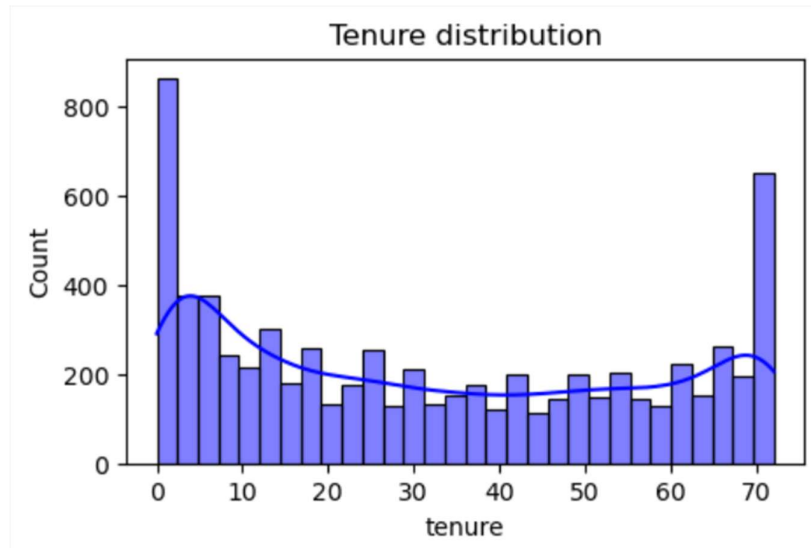
### ➤ Churn distribution



- Customer retention: A large proportion of customers continue to use services which indicate relatively strong retention.

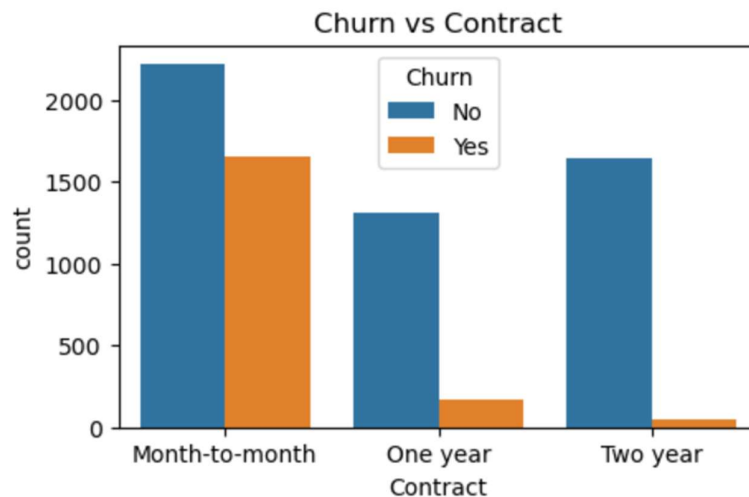
- Churn Rate: Significant number of customers has churned which suggest areas for improvement in customer satisfaction and service quality.

➤ **Tenure Distribution:**



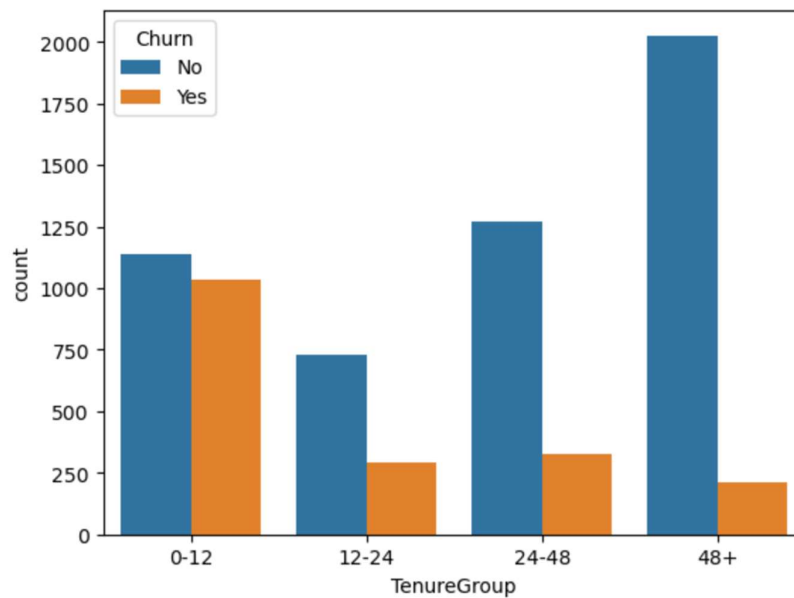
- High churn among new customers range 0-5 months, suggesting that many customers leave early.
- Stable retention among long term customers.

➤ **Churn Vs Contract**



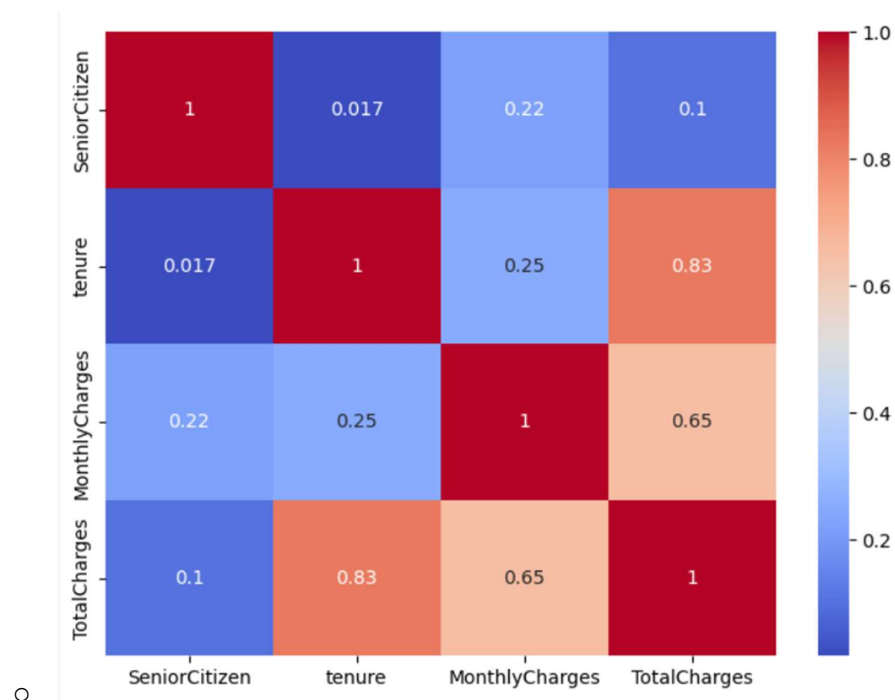
- Month-to-Month contract has high churn rate: It means that short term have the highest churn rate, indicating that customers with more flexible, short-term contracts are likely to churn.
- One year and Two year has lower churn: These two has lower churn rate suggesting that longer commitment contracts help in retention.

➤ Churn vs TenureGroupC



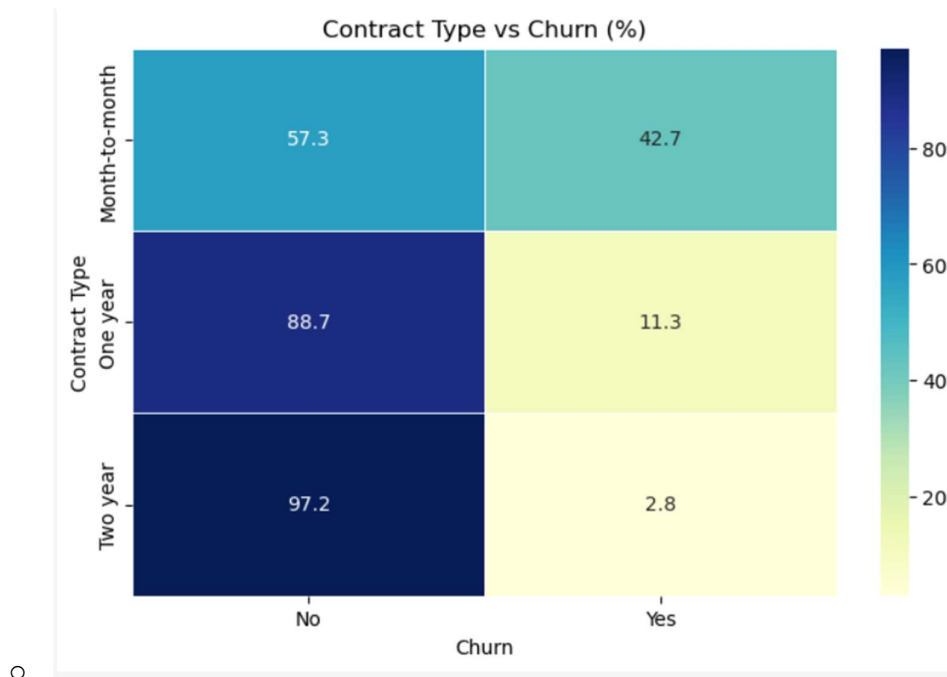
➤ Correlation Heatmap:

○ Numerical columns Heatmap:

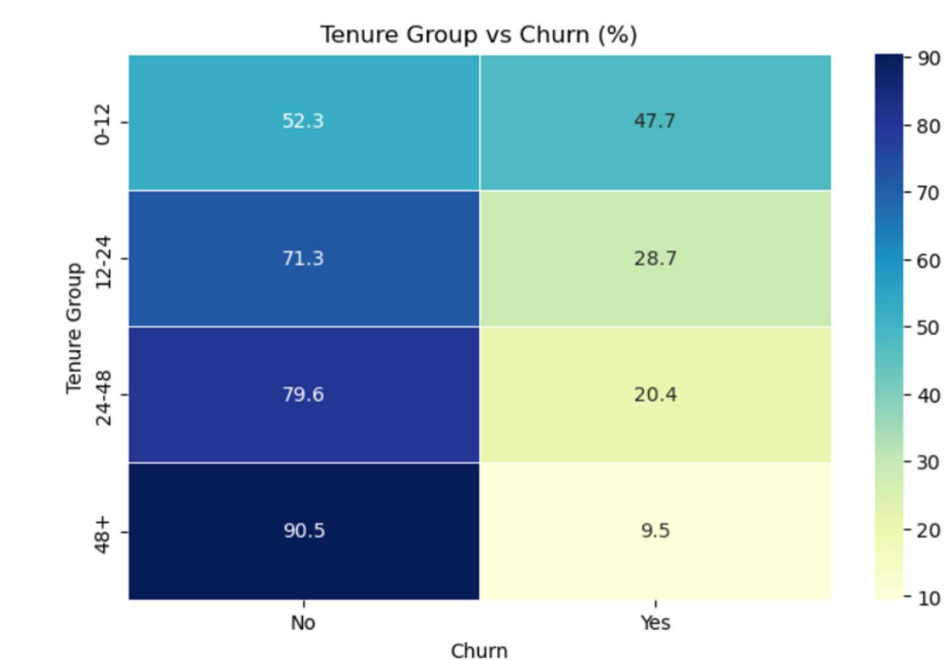


1. Tenure and total charges have a strong positive correlation. Long term customers contribute significantly to revenue.
2. Monthly charges and Total charges have moderate to strong positive correlation.
3. Tenure and monthlycharges show a weak positive correlation. Long term customers are not necessarily on expensive plans.

- **Churn vs Contract type Correlation**



- **Churn vs TenureGroup Correlation**



The analysis reveals that customer churn is highest during the first year of service. As customer tenure increases, the likelihood of churn decreases significantly. Customers who remain with the company for more than 48 months demonstrate the strongest retention rates, indicating that long-term customer relationships contribute positively to customer loyalty and business stability.

## Data Analysis using SQL

The python notebook is connect to PostgreSQL and loaded the cleaned dataframe into the database for SQL analysis.

- **Basic Operations**

- Total Customers:

|   | total_customers<br>bigint |
|---|---------------------------|
| 1 | 7043                      |

- Churn Distribution:

| Churn<br>text | customer_count<br>bigint |
|---------------|--------------------------|
| No            | 5174                     |
| Yes           | 1869                     |

- Churn Percentage:

| Churn<br>text | percentage<br>numeric |
|---------------|-----------------------|
| No            | 73.46                 |
| Yes           | 26.54                 |

- **Contract Analysis**

- Contract vs Churn

|   | Churn<br>text | Contract<br>text | total_customers<br>bigint |
|---|---------------|------------------|---------------------------|
| 1 | Yes           | Month-to-mon...  | 1655                      |
| 2 | No            | Month-to-mon...  | 2220                      |
| 3 | Yes           | One year         | 166                       |
| 4 | No            | One year         | 1307                      |
| 5 | Yes           | Two year         | 48                        |
| 6 | No            | Two year         | 1647                      |

- Churn rate by Contract

| Contract<br>text | churn_rate<br>numeric |
|------------------|-----------------------|
| One year         | 11.27                 |
| Month-to-mon...  | 42.71                 |
| Two year         | 2.83                  |

- **Revenue Analysis**

- Average Monthly charges by churn

|   | Churn<br>text | avg_monthly_charge<br>double precision |
|---|---------------|----------------------------------------|
| 1 | No            | 61.26512369539991                      |
| 2 | Yes           | 74.4413322632423                       |

- Average tenure by churn

|   | Churn<br>text | avg_tenure<br>numeric |
|---|---------------|-----------------------|
| 1 | No            | 37.57                 |
| 2 | Yes           | 17.98                 |

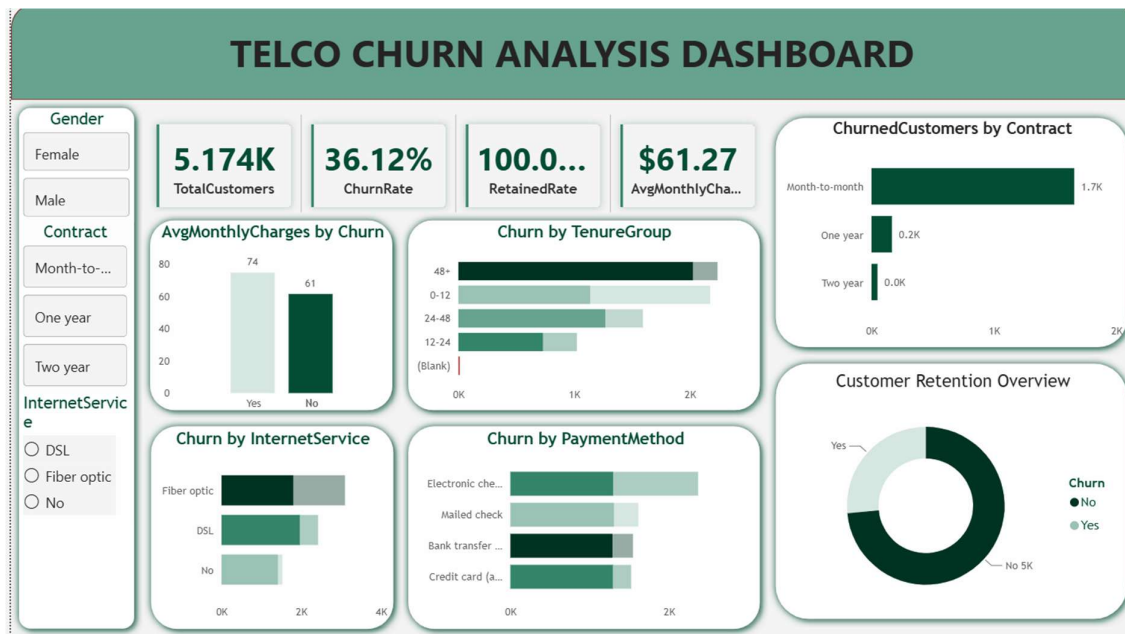
- Payment method by churn

|   | PaymentMethod<br>text     | Churn<br>text | customers<br>bigint |
|---|---------------------------|---------------|---------------------|
| 1 | Bank transfer (automat... | No            | 1286                |
| 2 | Bank transfer (automat... | Yes           | 258                 |
| 3 | Credit card (automatic)   | Yes           | 232                 |
| 4 | Credit card (automatic)   | No            | 1290                |
| 5 | Electronic check          | Yes           | 1071                |
| 6 | Electronic check          | No            | 1294                |
| 7 | Mailed check              | No            | 1304                |
| 8 | Mailed check              | Yes           | 308                 |

- Churn by internet service

|   | InternetService<br>text | churn_rate<br>numeric |
|---|-------------------------|-----------------------|
| 1 | Fiber optic             | 41.89                 |
| 2 | DSL                     | 18.96                 |
| 3 | No                      | 7.40                  |

## Dashboard using Power BI



## Business Recommendations

- **Improve Early Customer Engagement** – Focus retention efforts on customers within their first 12 months, as this group shows the highest churn risk.
- **Encourage Long-Term Contracts** – Promote one-year and two-year contract plans through discounts and incentives to reduce customer churn.
- **Retain High-Value Customers** – Develop loyalty rewards, personalized offers, and premium support for customers with high monthly spending.
- **Enhance Customer Support Services** – Strengthen technical support and customer service quality to improve customer satisfaction and retention.
- **Promote Service Bundles** – Encourage customers to subscribe to multiple services by offering bundled packages and special pricing.
- **Improve Customer On-boarding** – Provide guidance, tutorials, and proactive support during the initial months to build long-term customer relationships.
- **Use Data-Driven Marketing** – Leverage customer behaviour insights to deliver personalized promotions and retention-focused communication.